Constrained nonlinear optimization

Worked calculations and numerical verification

2105623 Optimization of Chemical Processes

November 16, 2026

Class sequence

0–15 min: direction, step size and feasibility

15–90 min: equality constraints and KKT with practice

90–100 min: break. 100–170 min: solver and sensitivity

170–180 min: assignment and final questions

A complete optimization calculation

1. State the objective, constraints and variable domains
2. Derive a candidate using a stated assumption
3. Substitute into every condition and calculate the value
4. Interpret optimality and the parameter range

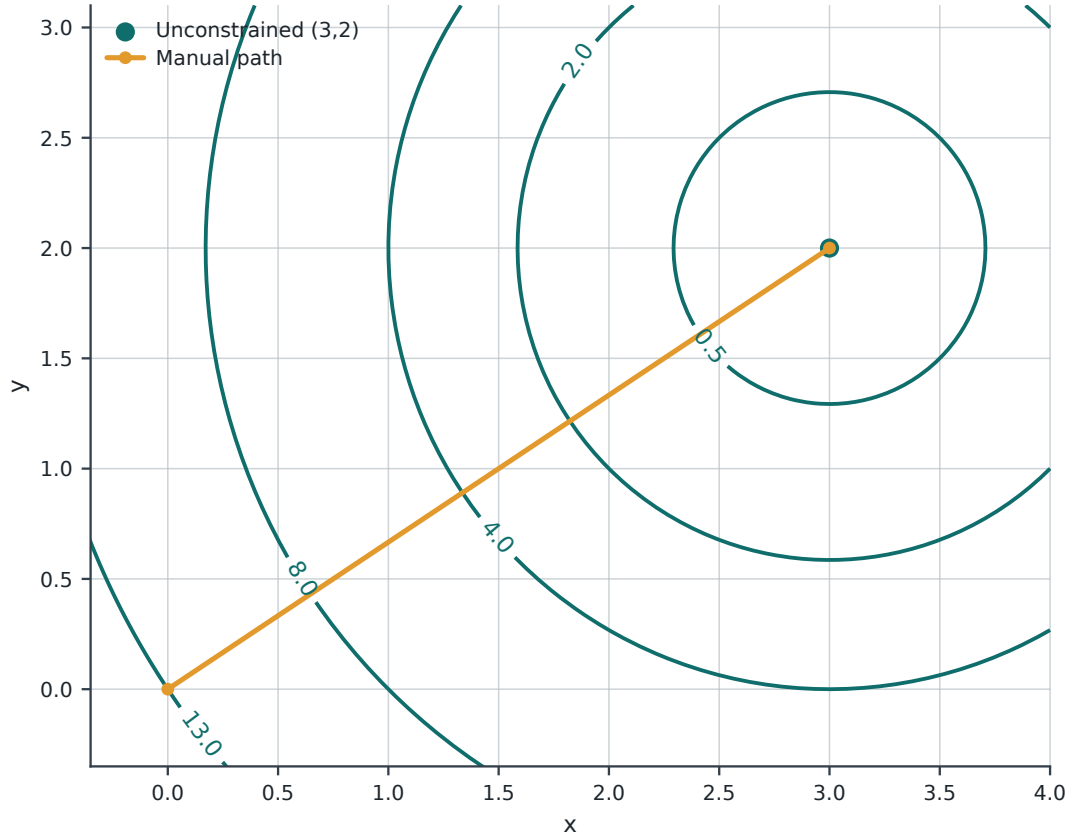
One step with exact line search

$f=(x-3)^2+(y-2)^2$, starting at $r_0=(0,0)$

$\nabla f(r_0)=(-6,-4)$, so $p_0=(6,4)$

$\varphi(\alpha)=13-52\alpha+52\alpha^2$. Set $\varphi'=-52+104\alpha=0$

$\alpha^*=1/2$ gives $r_1=(3,2)$, with f falling from 13 to 0



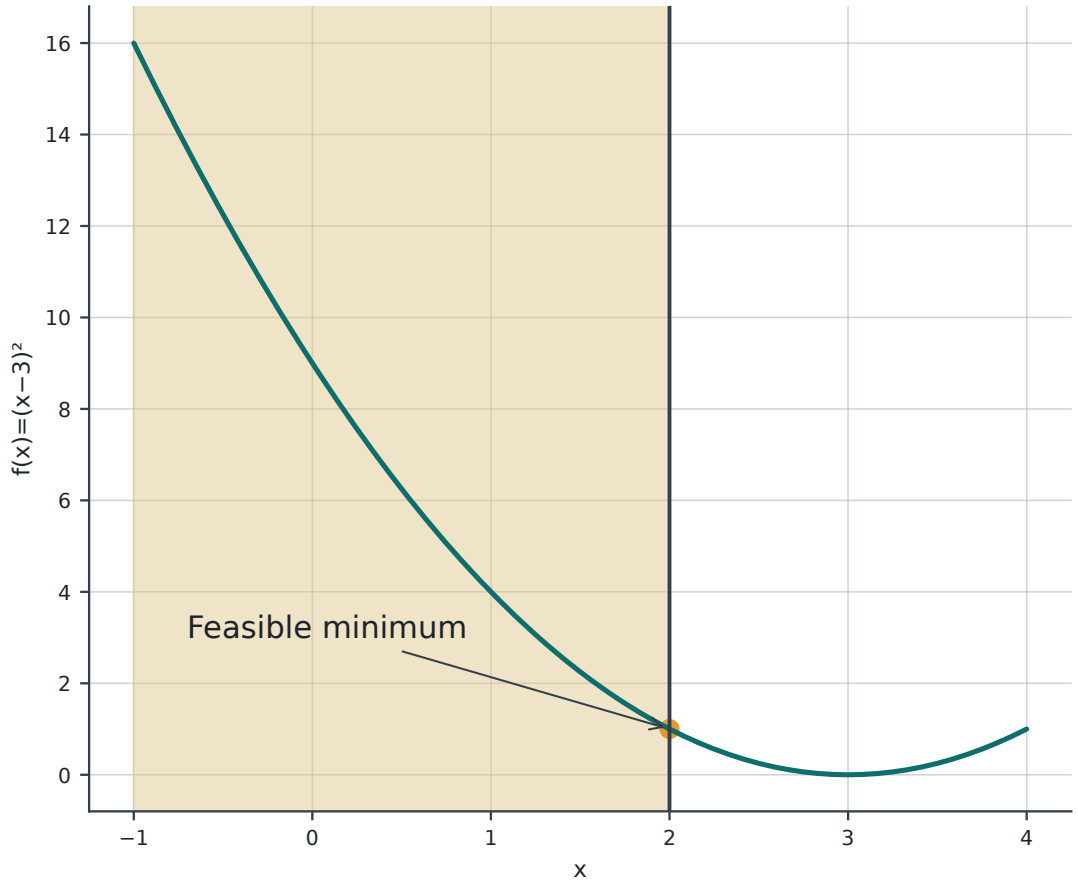
A boundary can block descent

$\min f(x)=(x-3)^2$ subject to $x \le 2$

$f'(x)=2(x-3)$, so $f'(2)=2(-1)=-2$

At $x=2$, descent needs $d>0$ but feasibility needs $d \le 0$

The constrained minimum is $x^*=2$ with $f^*=1$



The sign convention for this class

min $f(r)$ with $g_i(r) \leq 0$ and $h_j(r) = 0$

$L = f + \Sigma \lambda_i g_i + \Sigma \nu_j h_j$

Inequality multipliers: $\lambda_i \geq 0$

Equality multipliers: ν_j can have either sign

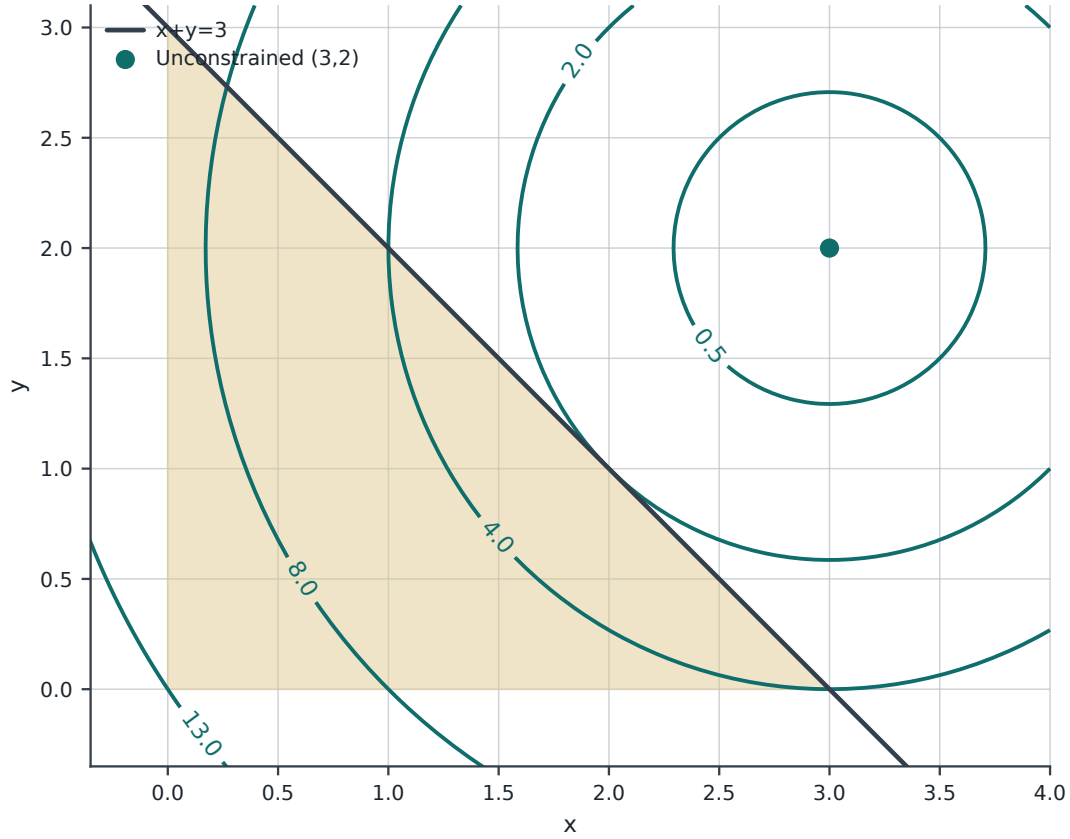
The main production example

$f(x,y)=(x-3)^2+(y-2)^2$

The unconstrained target is (3,2)

Available capacity is $x+y\leq 3$ with $x,y\geq 0$

We will find the best feasible point and verify it



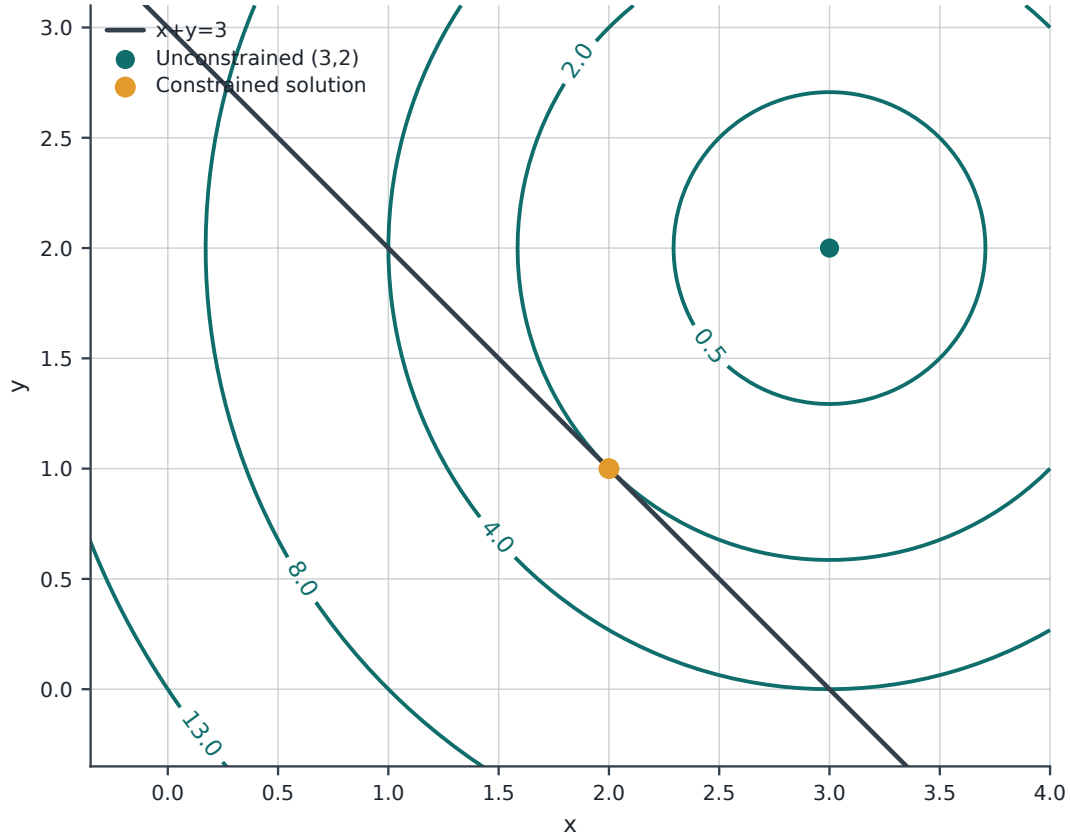
Equality step 1 One feasible coordinate

First impose x+y=3 exactly

Substitute y=3−x into the objective

$f(x,3-x)=(x-3)^2+(1-x)^2$

Now minimize a function of x alone



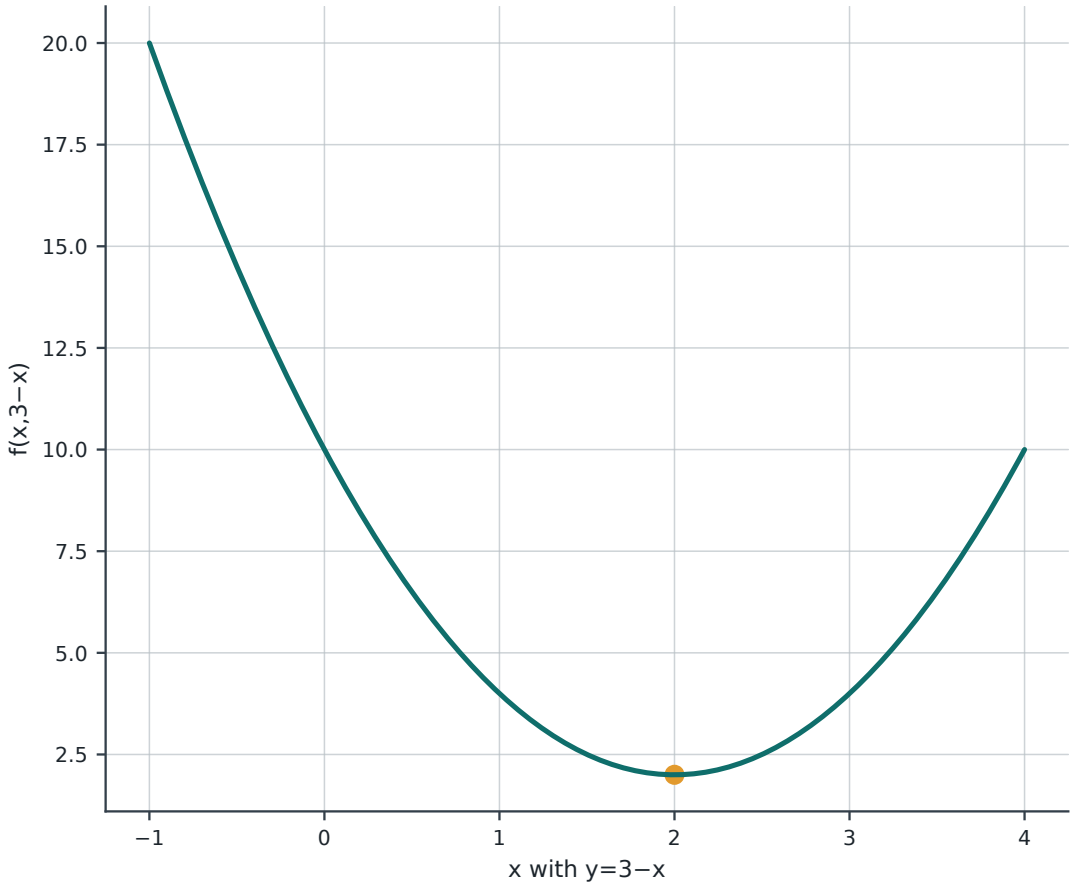
Equality step 2 Differentiate the section

d f(x,3−x)/dx = 2(x−3)+2(x−1)

4x−8=0 gives x=2

Then y=3−2=1 and f=1+1=2

The second derivative is 4>0



Equality step 3 Lagrange equations

$h=x+y-3$ and $L=f+\nu h$

$\partial L/\partial x = 2(x-3)+\nu = 0$

$\partial L/\partial y = 2(y-2)+\nu = 0$

The third equation is $x+y=3$

Equality step 4 Solve the multiplier

$x=3-\nu/2$ and $y=2-\nu/2$

Substitute into $x+y=3$: $5-\nu=3$

Hence $\nu=2$, $x=2$ and $y=1$

Substitution and Lagrange give the same result

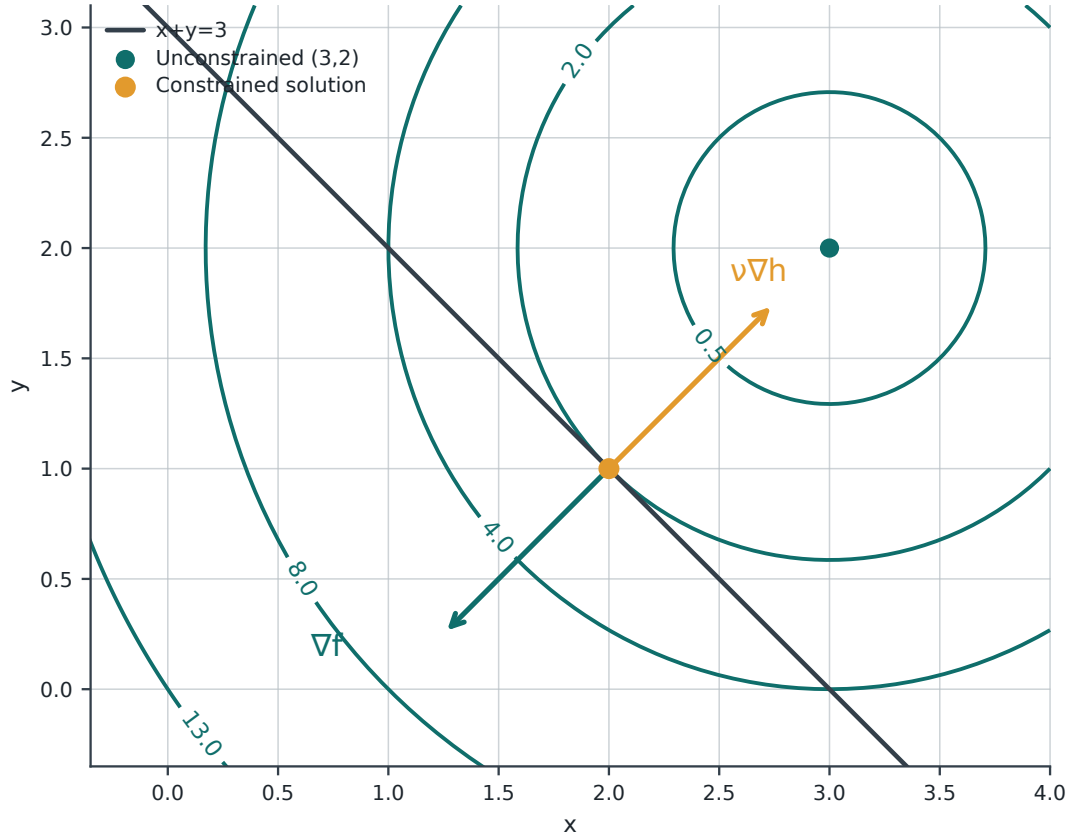
Equality step 5 Gradient balance

At (2,1), $\nabla f=(-2,-2)$

The constraint normal is $\nabla h=(1,1)$

$\nu\nabla h=2(1,1)=(2,2)$

$\nabla f+\nu\nabla h=(0,0)$



The four KKT checks

Stationarity: $\nabla f + \Sigma \lambda_i \nabla g_i + \Sigma \nu_j \nabla h_j = 0$

Primal feasibility: $g_i \leq 0$ and $h_j = 0$

Dual feasibility: $\lambda_i \geq 0$

Complementarity: $\lambda_i g_i = 0$ for every inequality

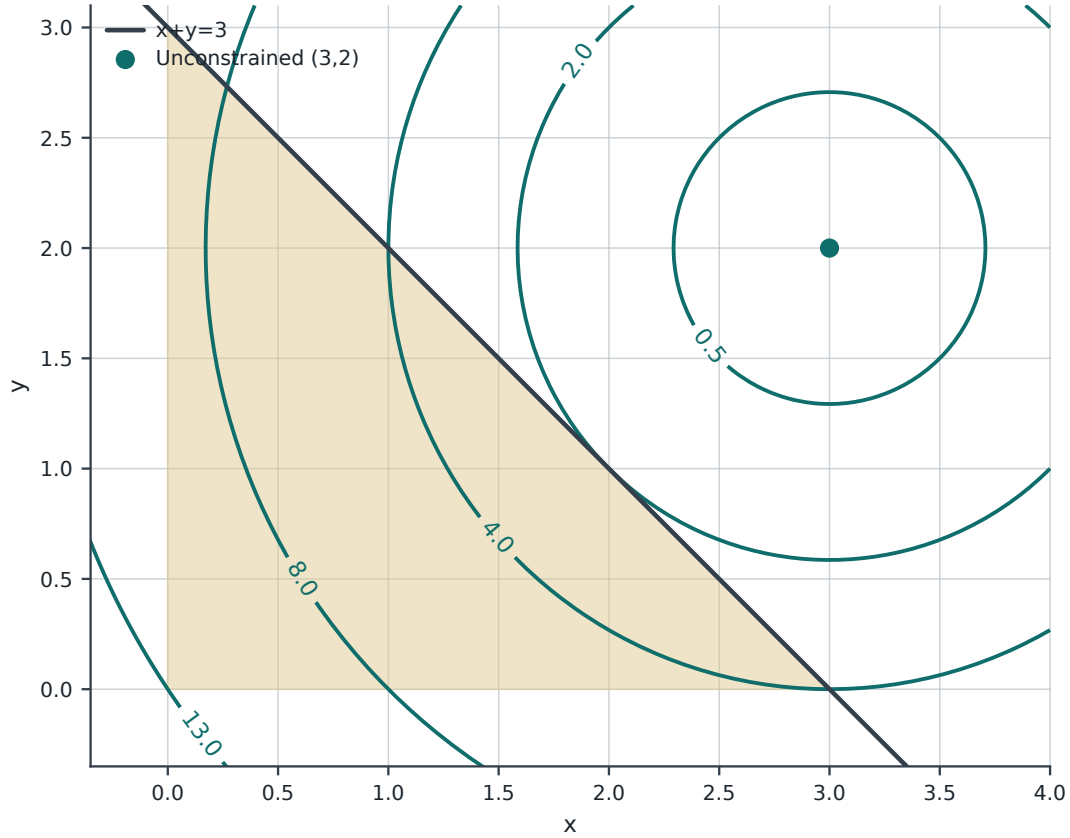
Inequality step 1 All constraints

$\min f=(x-3)^2+(y-2)^2$

$g_1=x+y-3\leq 0$

$g_2=-x\leq 0$ and $g_3=-y\leq 0$

$L=f+\lambda_1(x+y-3)-\lambda_2 x-\lambda_3 y$



Inequality step 2 The unconstrained candidate

$\nabla f = 0$ gives $x = 3$ and $y = 2$

Capacity check: $g_1 = 3 + 2 - 3 = 2 > 0$

The point has $f = 0$ but violates the capacity constraint

A low objective cannot compensate for infeasibility

Inequality step 3 An active-set assumption

Assume $x+y=3$ and $x>0$, $y>0$

Then $g_2<0$ and $g_3<0$

Complementarity gives $\lambda_2=\lambda_3=0$

We will check the assumed signs after solving

Inequality step 4 Stationarity

$2(x-3)+\lambda_1-\lambda_2=0$

$2(y-2)+\lambda_1-\lambda_3=0$

With $\lambda_2=\lambda_3=0$: $x=3-\lambda_1/2$

Similarly, $y=2-\lambda_1/2$

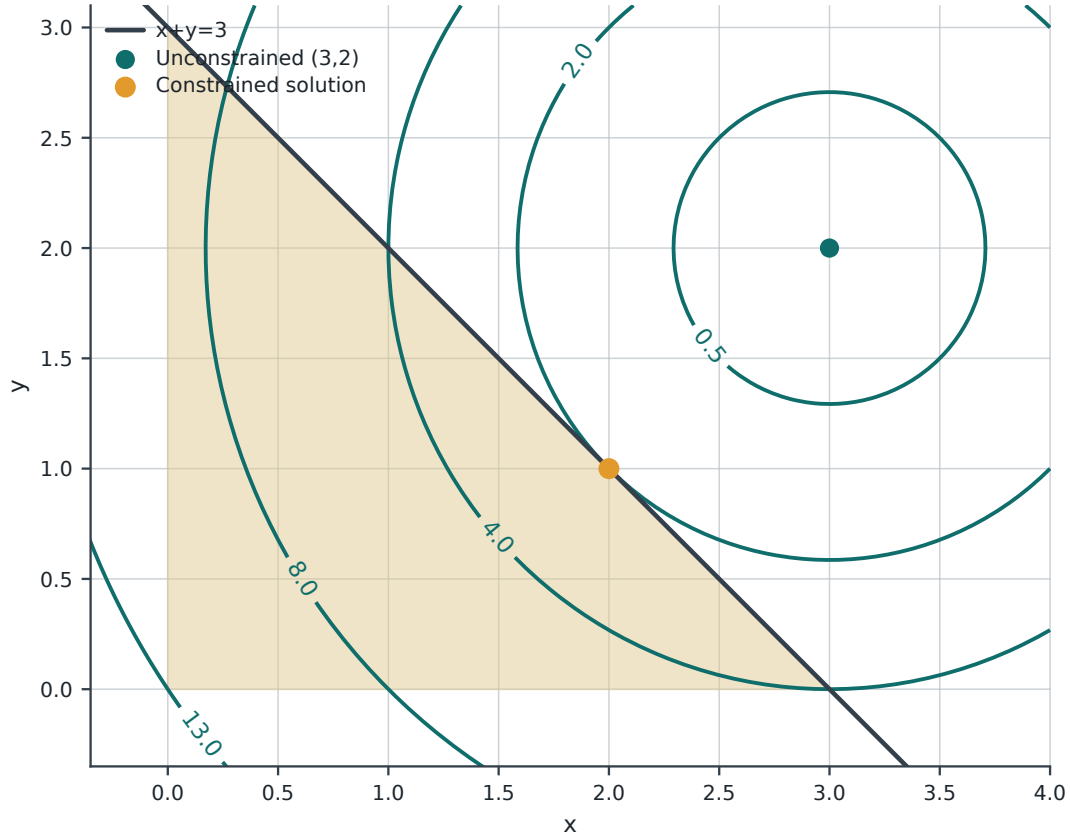
Inequality step 5 Solve the active equation

$x+y=3$ gives $3-\lambda_1/2+2-\lambda_1/2=3$

$5-\lambda_1=3$, hence $\lambda_1=2$

$x=3-1=2$ and $y=2-1=1$

Both coordinates are positive, as assumed



Inequality step 6 Feasibility checks

At $(2,1)$: $g=(0,-2,-1)$

Every $g_i \leq 0$, including both lower bounds

The multipliers are $\lambda=(2,0,0)$

Every $\lambda_i \geq 0$

Inequality step 7 Complementarity

$\lambda_1 g_1 = 2(0) = 0$

$\lambda_2 g_2 = 0(-2) = 0$

$\lambda_3 g_3 = 0(-1) = 0$

Each multiplier pairs with its own constraint

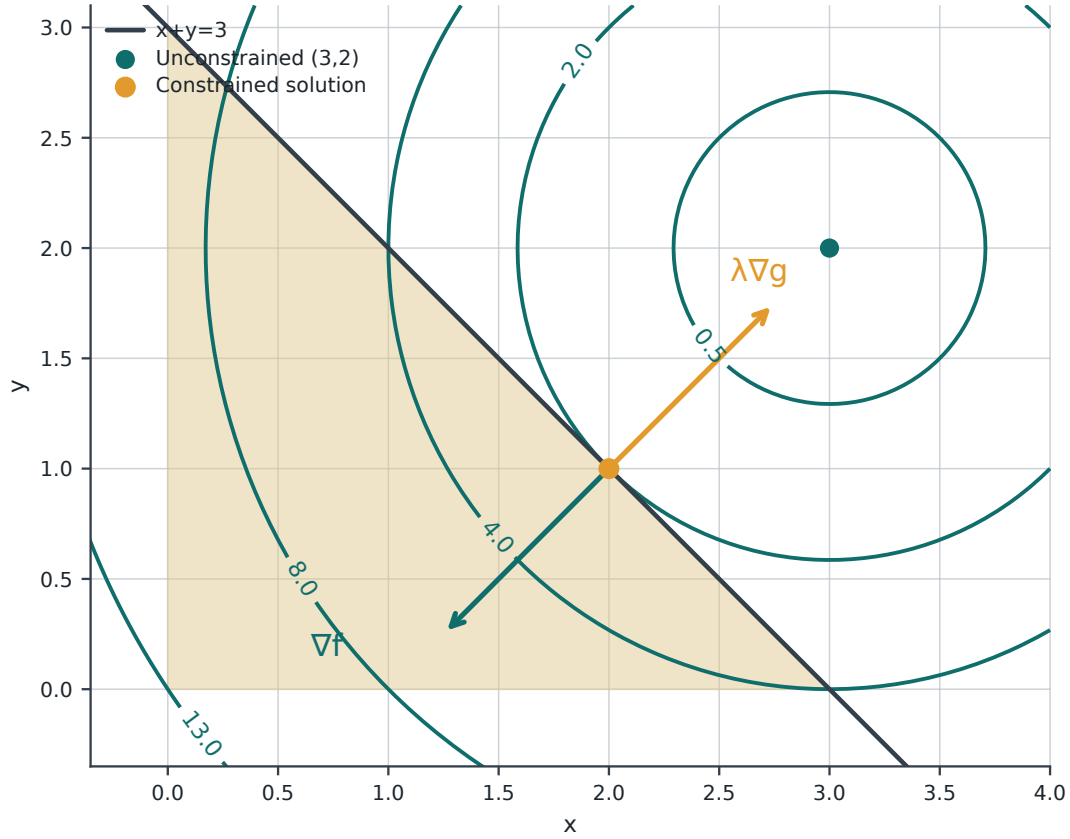
Inequality step 8 Stationarity and value

$\nabla f(2,1)=(-2,-2)$

$\lambda_1\nabla g_1+\lambda_2\nabla g_2+\lambda_3\nabla g_3=(2,2)$

The stationarity residual is (0,0)

$f(2,1)=(-1)^2+(-1)^2=2$



Inequality step 9 Global optimality

The Hessian of f is 2I, which is positive definite

All constraints are affine, so the feasible set is convex

A feasible KKT certificate proves global optimality here

Strict convexity gives the unique solution (2,1)

An active constraint can have zero price

$\min x^2$ subject to $x \leq 0$

At $x^*=0$, the bound is active

Stationarity gives $2(0)+\lambda=0$, so $\lambda=0$

Activity alone does not imply a positive multiplier

When KKT is necessary

Suitable constraint qualifications support KKT necessity

For affine constraints, inspect the active normals

In our example, the only active normal is (1,1)

A verified KKT point in a convex problem gives sufficiency

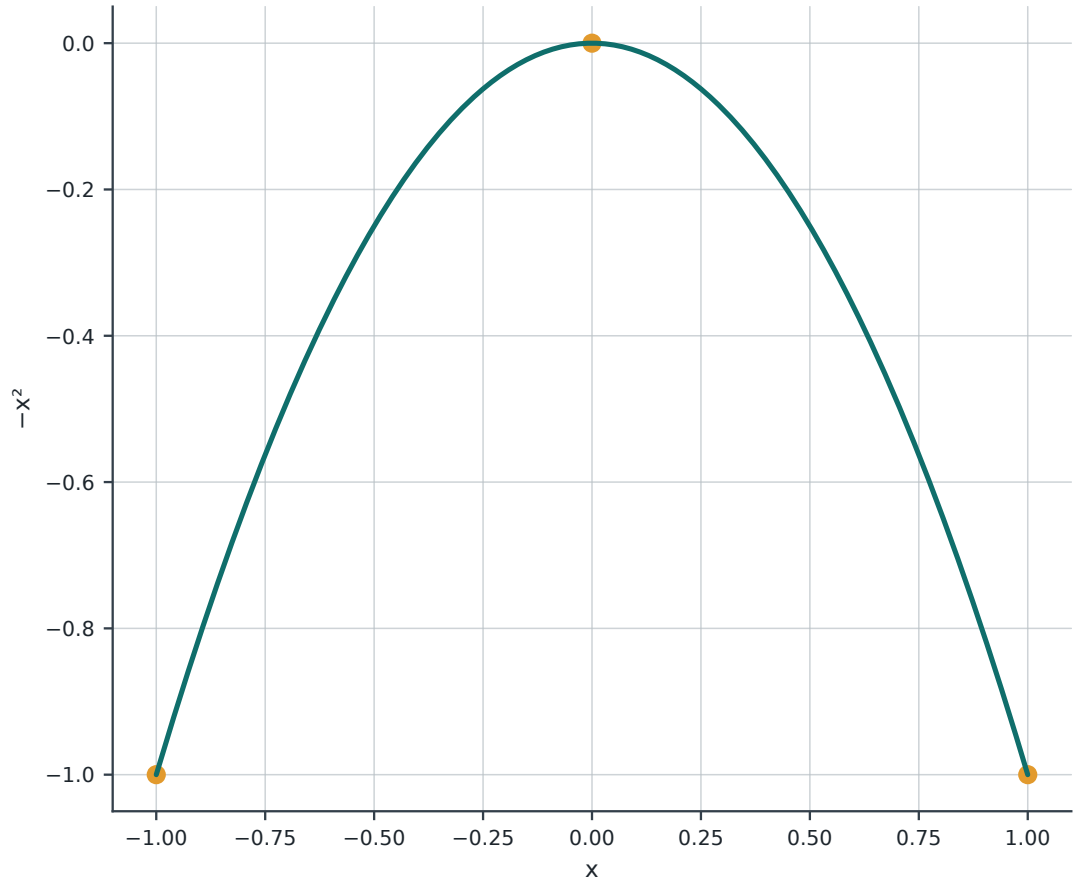
A nonconvex KKT point can be a maximum

min $-x^2$ subject to $-1 \leq x \leq 1$

At x=0, both bounds are inactive and λ=0

∇f=0 satisfies stationarity, yet f(0)=0

At x=±1, f=−1, so x=0 is not a minimum



Practice A one-variable KKT problem

$\min (x-4)^2$ subject to $x \le 1$

1. Write g and L using today’s convention

2. Test $\lambda=0$, then test the active bound

3. Verify KKT and calculate f*

Practice solution Two cases

If $\lambda=0$, stationarity gives $x=4$, which is infeasible

If $x=1$, then $2(1-4)+\lambda=0$ gives $\lambda=6$

$g=0$, $\lambda=6\geq 0$ and $\lambda g=0$

$f^*=9$. Convexity certifies the unique global minimum

Break

10 minutes

Next: a numerical solution and an independent audit

A numerical solution needs an audit

State the mathematical model and solver interface

Run the numerical method and read its status

Recompute the objective and constraint values

Check multipliers, stationarity and complementarity

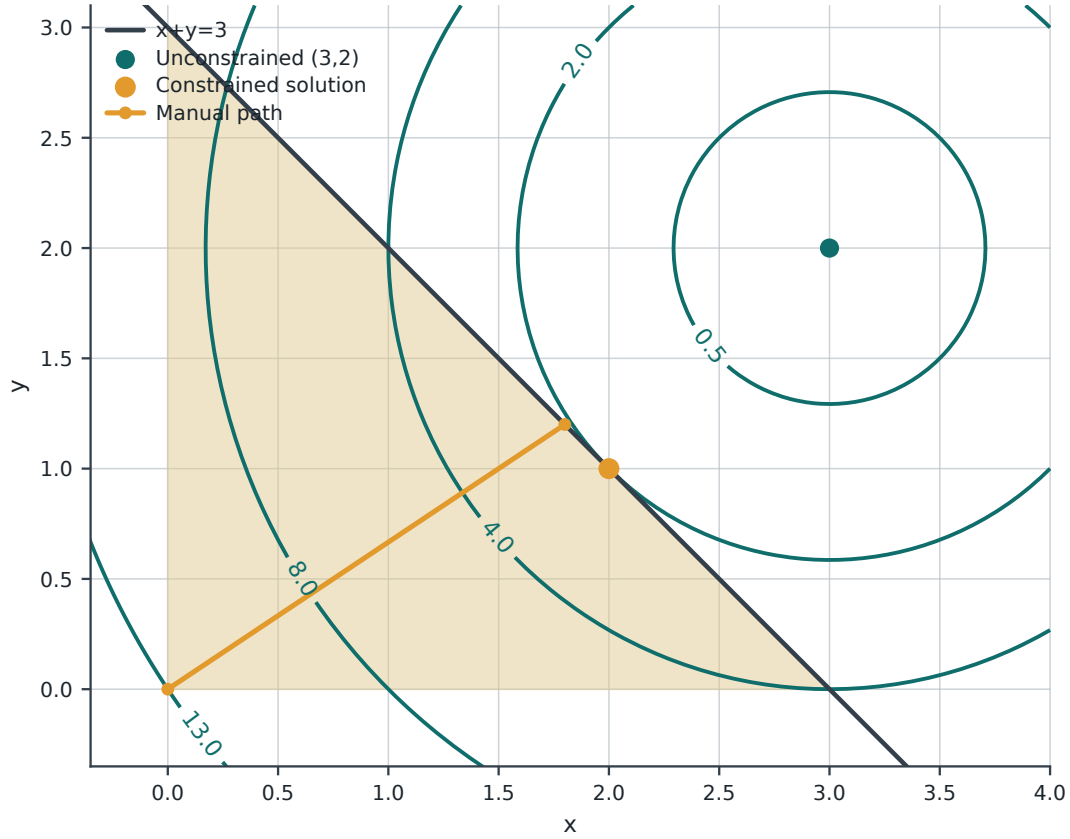
A feasible first step from the origin

At $r_0=(0,0)$, choose $p=(6,4)$

Capacity requires $10\alpha \le 3$, so $0 \le \alpha \le 0.3$

At $\alpha=0.3$, $r_1=(1.8,1.2)$

$f(r_1)=1.2^2+0.8^2=2.08$



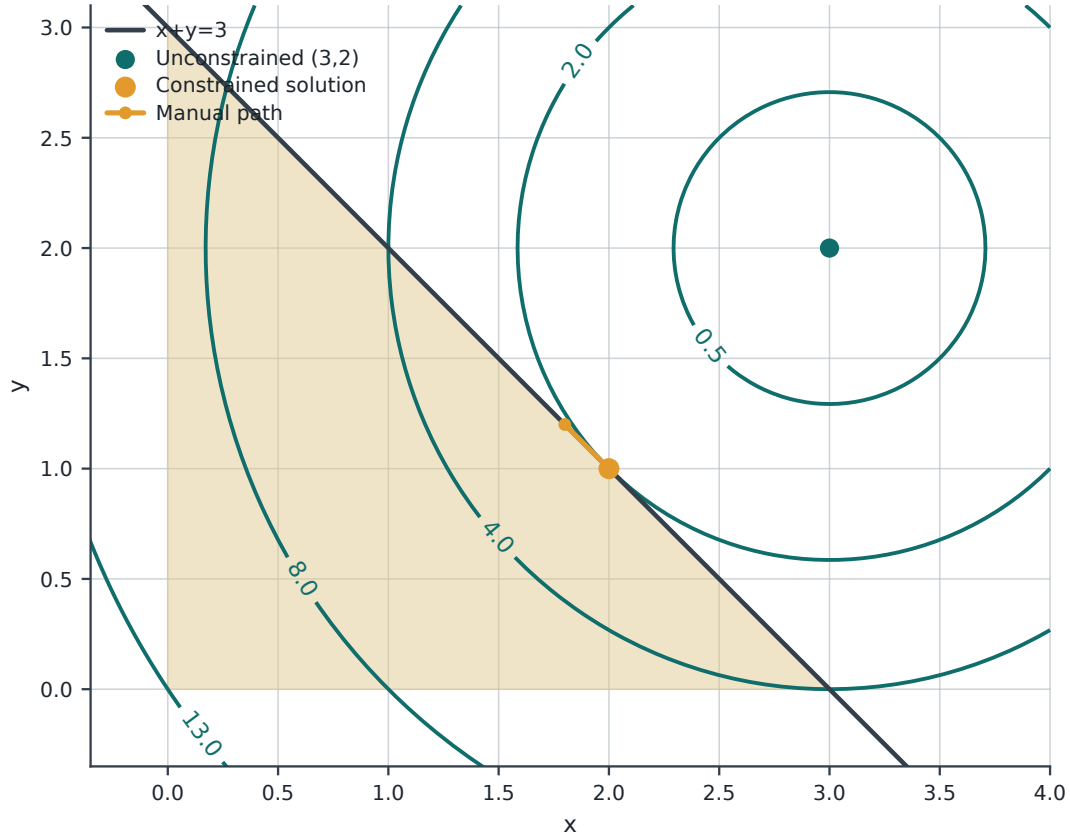
A correction along the boundary

At $r_1=(1.8,1.2)$, take $d=(0.2,-0.2)$

$d_x+d_y=0$ keeps $x+y=3$

$\nabla f(r_1)^T d=(-2.4)(0.2)+(-1.6)(-0.2)=-0.16$

At step 1: $r_2=(2,1)$, $f(r_2)=2$



Python step 1 Objective and gradient

```
def objective(r):

    x, y = r

    return (x-3)**2 + (y-2)**2

def gradient(r): return 2*(np.asarray(r)-[3,2])
```

Python step 2 The inequality sign

Mathematics: $g = x + y - 3 \leq 0$

SLSQP dictionary interface: $c = 3 - x - y \geq 0$

```
constraint={"type":"ineq", "fun":lambda r:3-r.sum()}
```

```
bounds=[(0,None),(0,None)]
```

Python step 3 Solve and read status

```
result=minimize(objective, [0.5,0.5], jac=gradient,
    method="SLSQP", bounds=bounds, constraints=[constraint],
    options={"ftol":1e-12, "maxiter":200})
```

Read result.success, result.message, result.x and result.fun

Python step 4 Recompute the result

Expected point: $x \approx 2$, $y \approx 1$ and $f \approx 2$

Capacity residual: $x + y - 3 \approx 0$

Bound residuals: $-x \approx -2$ and $-y \approx -1$

A successful run should agree with the analytic certificate

Python step 5 Obtain the multipliers

At this solution, $x>0$ and $y>0$ imply $\lambda_2=\lambda_3=0$

x stationarity gives $\lambda_1=2(3-x)$

y stationarity gives $\lambda_1=2(2-y)$

At (2,1), both equations give $\lambda_1=2$

Python step 6 Four residuals

Primal: $\max(0, \max_i g_i)$

Dual: $\max(0, \max_i(-\lambda_i))$

Stationarity: $\|\nabla f + \Sigma \lambda_i \nabla g_i\|_\infty$

Complementarity: $\max_i |\lambda_i g_i|$

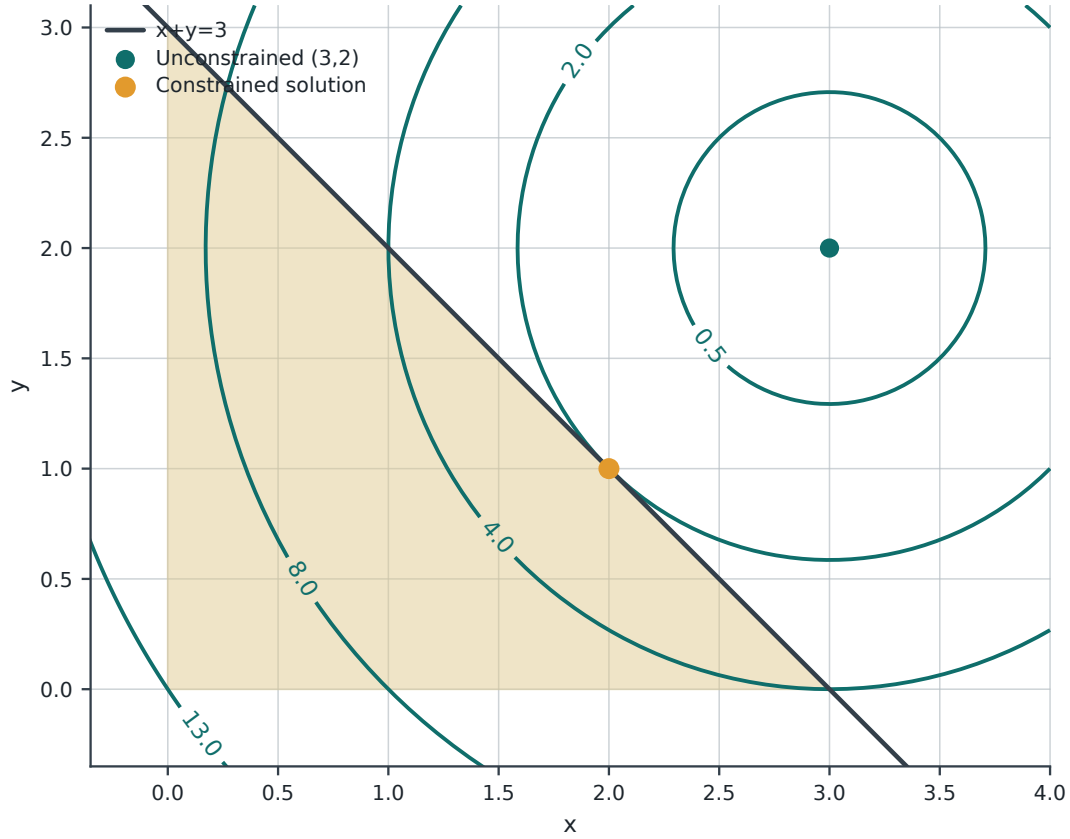
Capacity step 1 A changing right-hand side

Now let $x+y \le b$ with $x,y \ge 0$ and $b \ge 0$

In the middle regime, assume $x+y=b$ and $x,y>0$

Stationarity gives $x=3-\lambda_1/2$ and $y=2-\lambda_1/2$

Substitute: $5-\lambda_1=b$, so $\lambda_1=5-b$



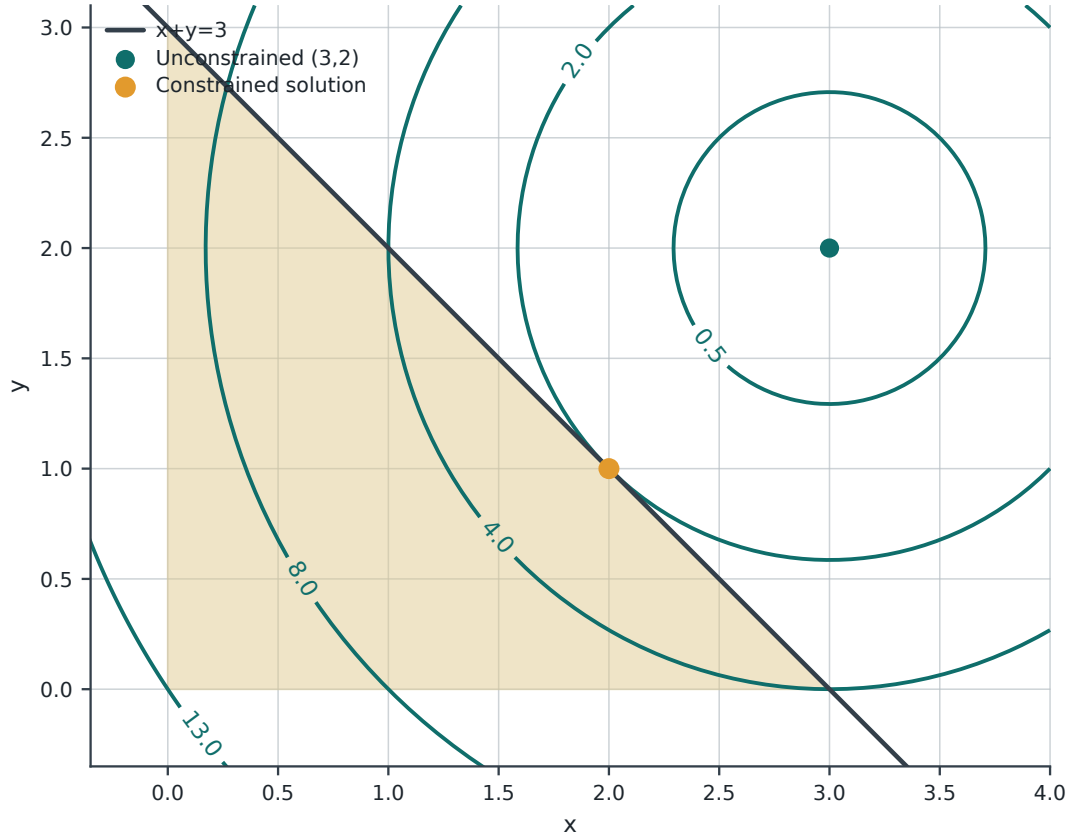
Capacity step 2 The valid interval

$x=(b+1)/2$ and $y=(b-1)/2$

Nonnegative y requires $b \geq 1$

Nonnegative $\lambda_1=5-b$ requires $b \leq 5$

The same KKT formula is valid on $1 \leq b \leq 5$



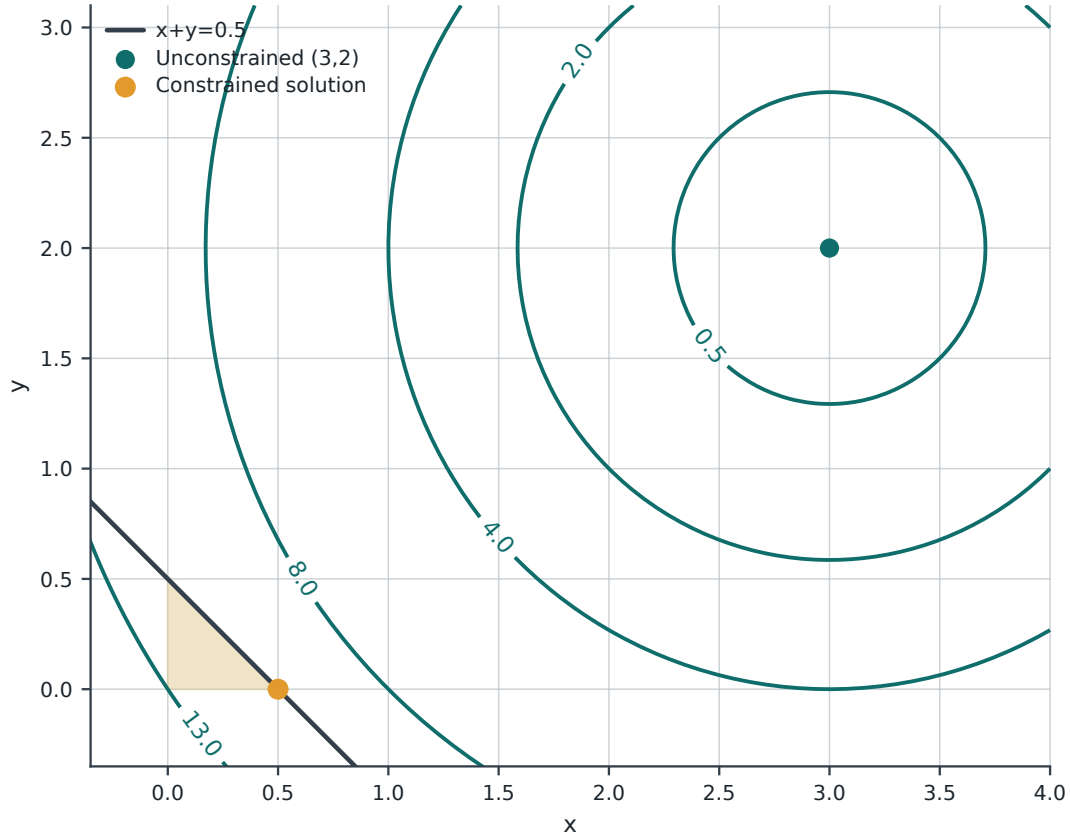
Capacity step 3 Too little capacity

For $0 \leq b < 1$, the middle formula would give $y < 0$

Try $y=0$ and $x=b$

Stationarity gives $\lambda_1 = 6 - 2b$ and $\lambda_3 = 2 - 2b$

$V(b) = (b-3)^2 + 4$, with $\lambda_2 = 0$ a valid choice



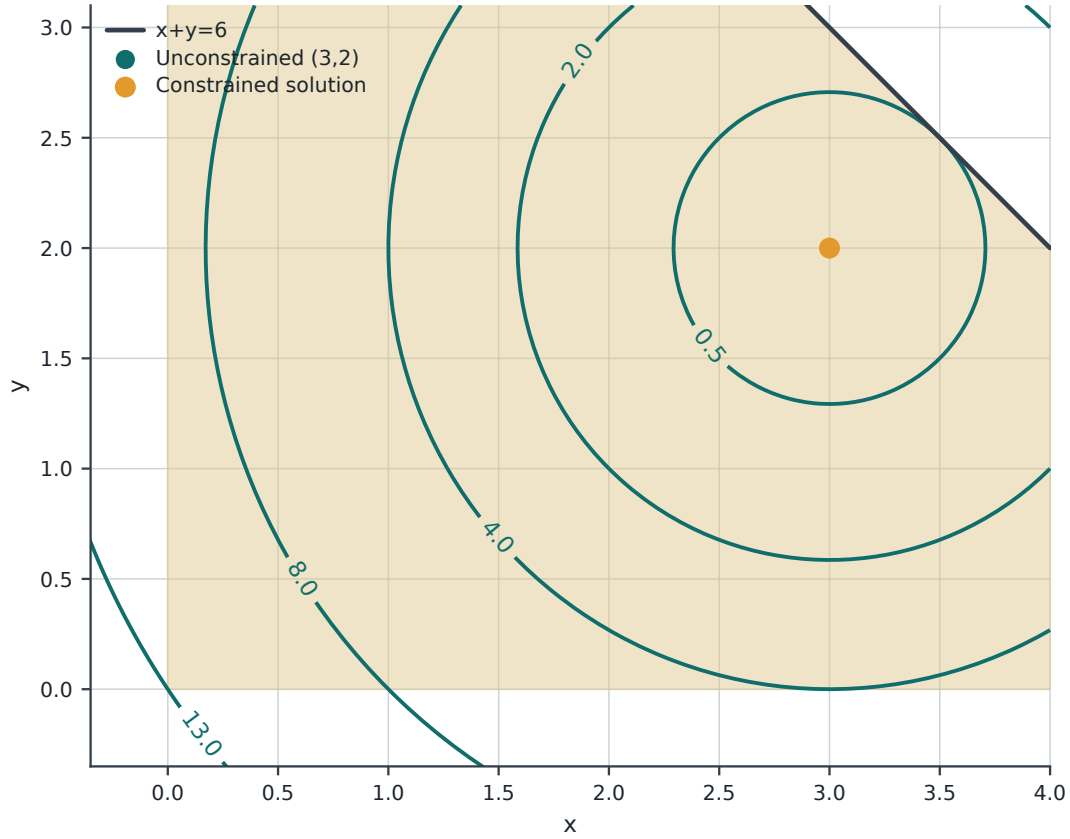
Capacity step 4 Enough capacity

For b>5, the target (3,2) is feasible

The capacity constraint is slack and $\lambda_1=0$

The optimum value is V(b)=0

At b=5 the capacity is active with zero price



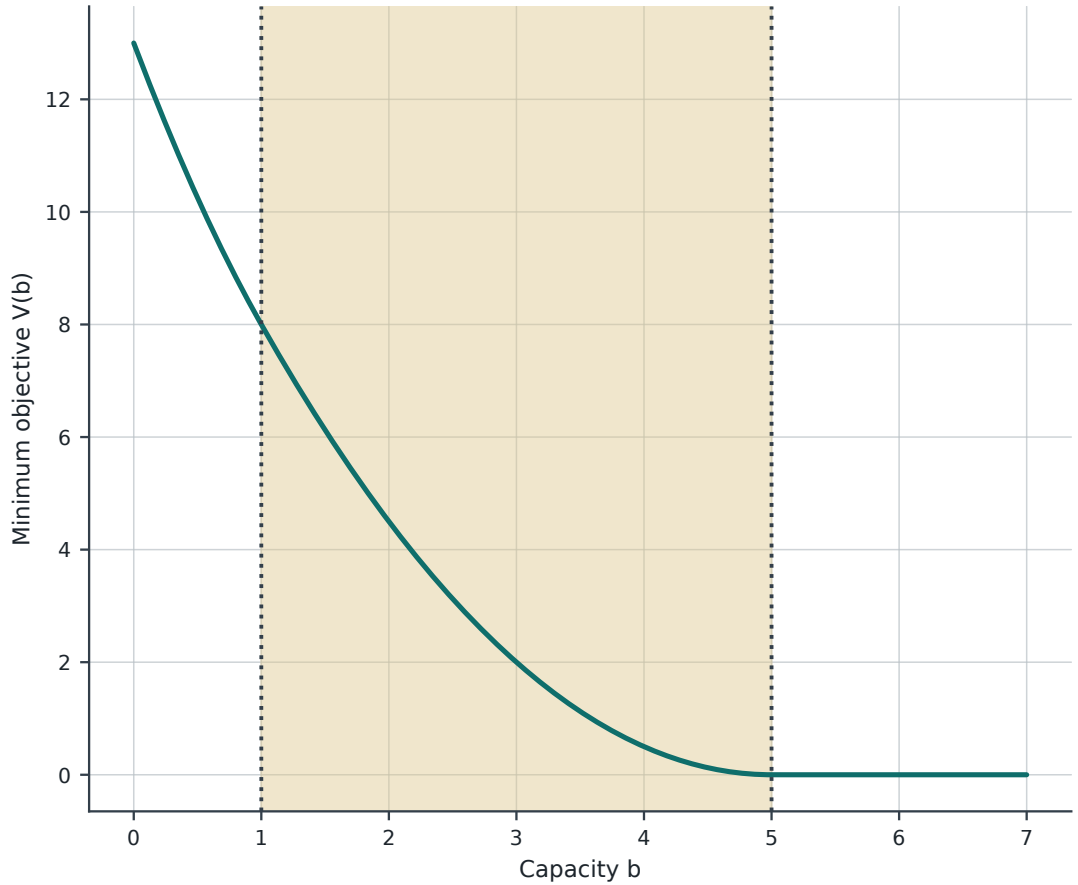
Capacity step 5 The complete value function

$0 \le b \le 1$: $V(b) = (b-3)^2 + 4$

$1 \le b \le 5$: $V(b) = (b-5)^2/2$

$b \ge 5$: $V(b) = 0$

The active set changes at $b=1$ and $b=5$



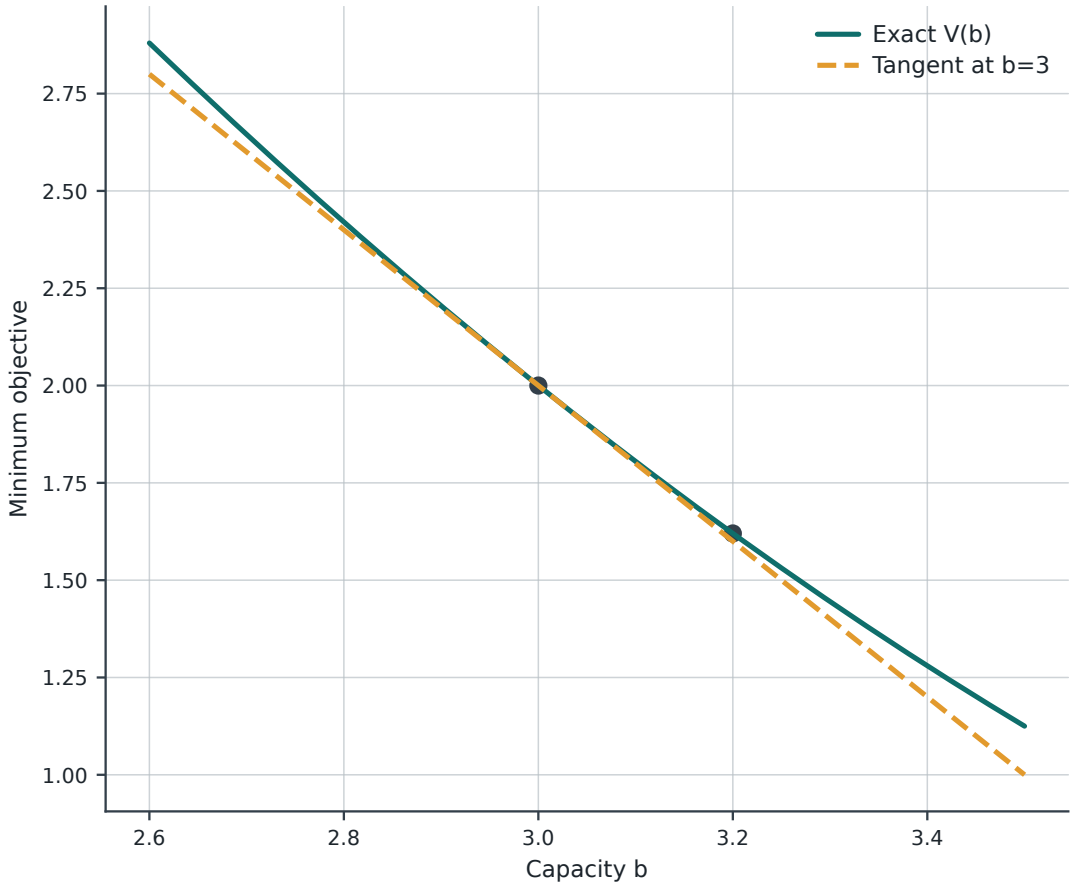
Sensitivity step 1 Exact and linear changes

At $b=3$: $V=2$ and $\lambda_1=2$

Increase capacity by $\Delta b=0.2$

First-order prediction: $\Delta V \approx -\lambda_1 \Delta b = -0.4$

The predicted value is $2-0.4=1.6$



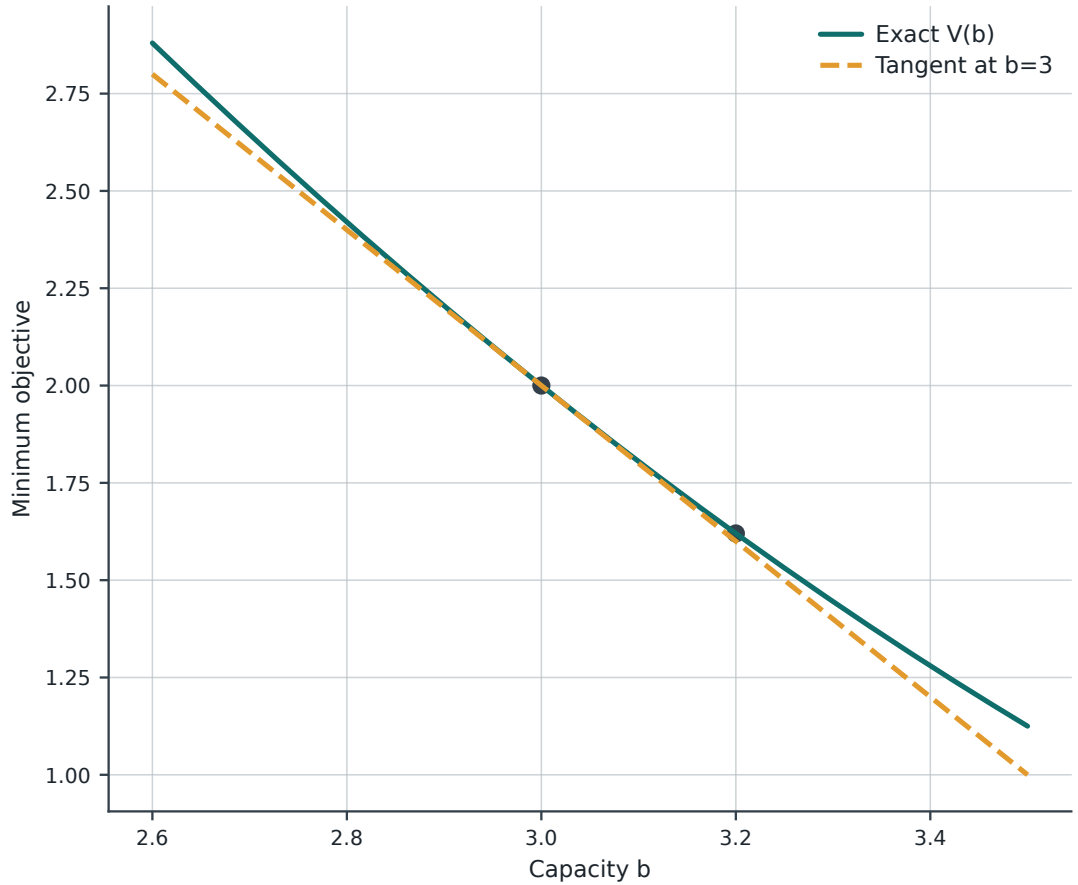
Sensitivity step 2 Reoptimize and compare

At b=3.2: x=2.1 and y=1.1

$V=(2.1-3)^2+(1.1-2)^2=0.81+0.81=1.62$

Exact $\Delta V=1.62-2=-0.38$

Linear prediction overstates the reduction by 0.02



Demo 1 The capacity slider

Move b through 0.5, 1, 3, 5 and 6

Before moving, predict which constraints are active

Compare point, objective and multiplier values

The display checks all four KKT residuals

Demo 2 Several numerical starts

Run the Python example from (0.5,0.5), (3,2), (0,0), (4,0)

Some starting points violate the capacity constraint

Compare the returned points and independent residuals

Multiple starts support consistency. Convexity supplies the global proof

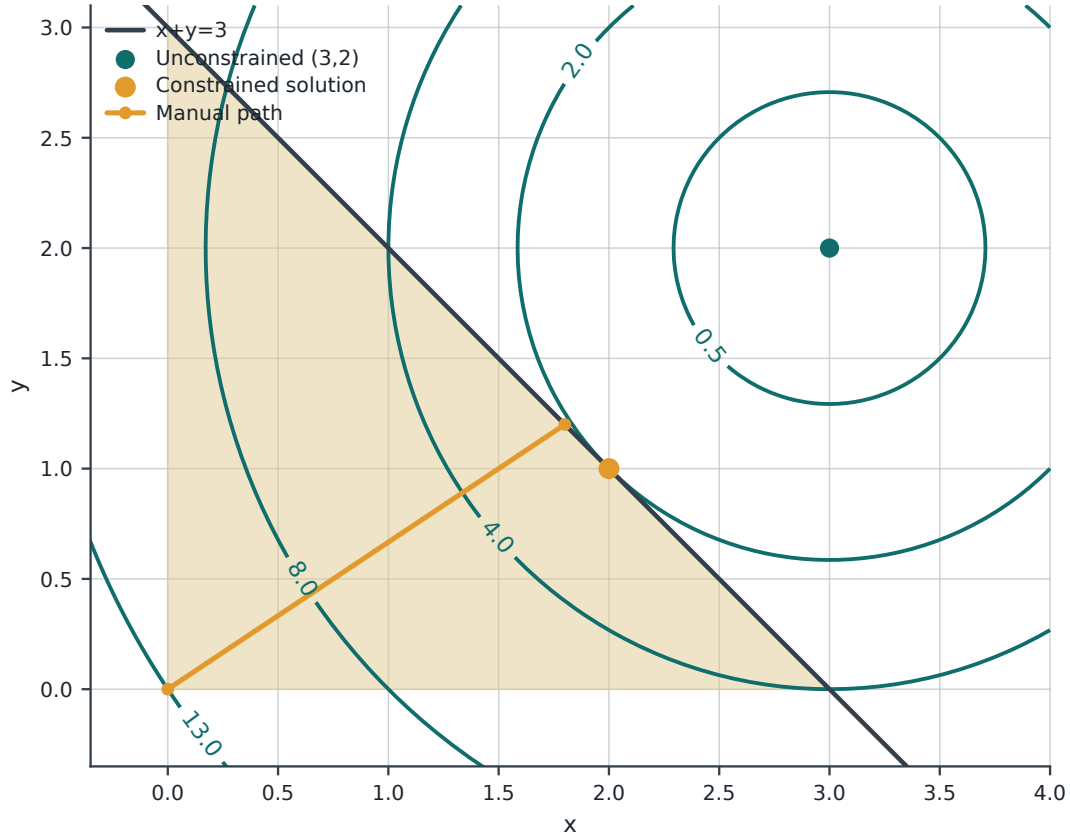
Demo 3 A feasible manual path

Compare α=0.1, 0.3 and 0.5 from r_0=(0,0), p=(6,4)

For each trial calculate x+y, f and feasibility

Only α≤0.3 respects capacity b=3

After reaching the boundary, change the direction



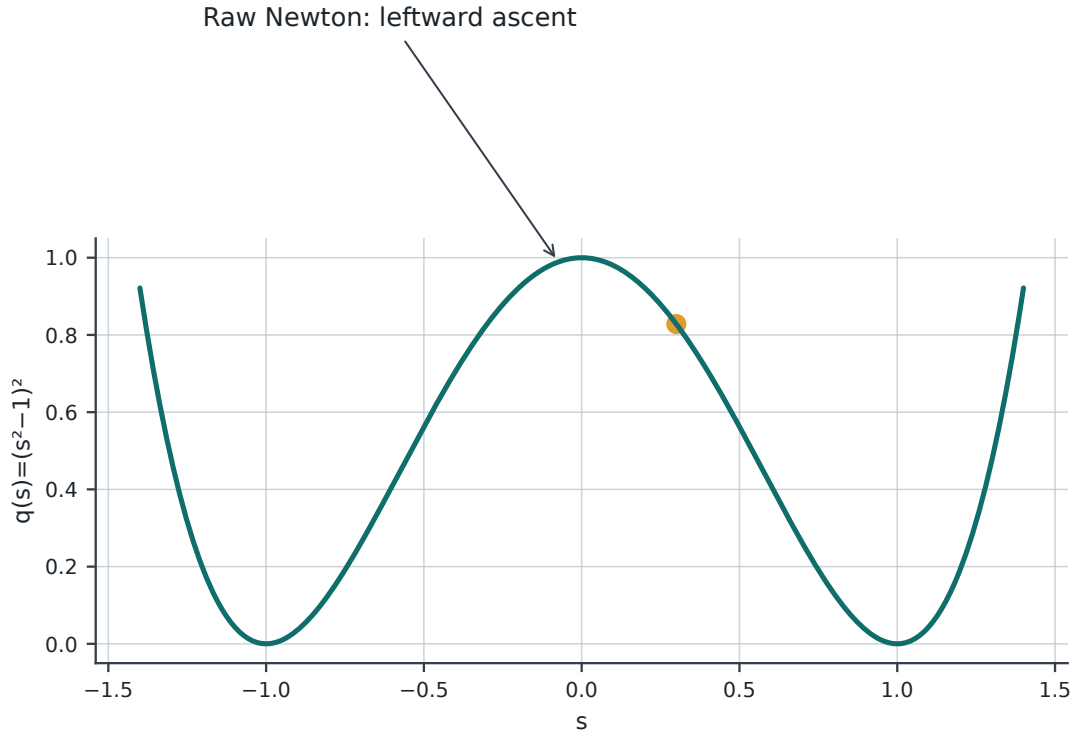
Demo 4 A failed Newton direction

$q(s)=(s^2-1)^2$ at $s=0.3$

$q'=-1.092$ and $q''=-2.92$

Raw Newton gives $p\approx-0.37397$ and $q'p>0$

A negative-gradient safeguard gives a descent direction



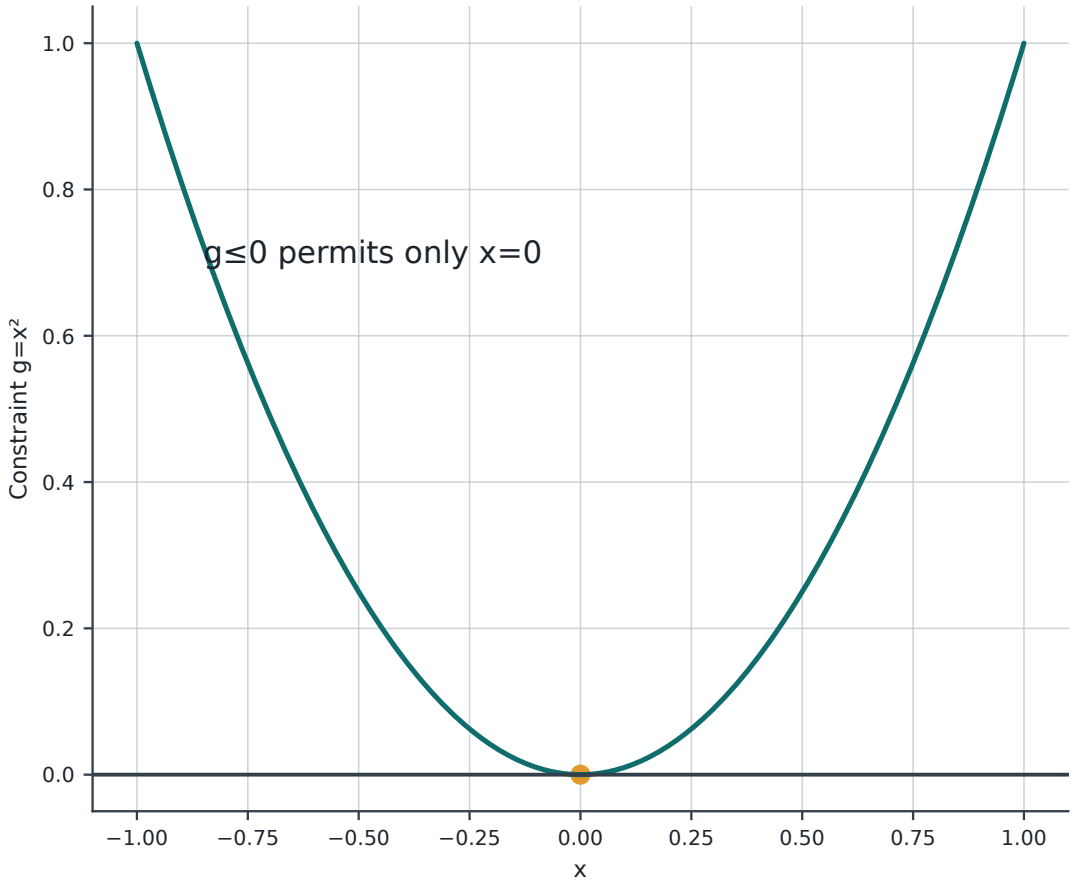
Demo 5 A missing regularity condition

min x subject to $x^2 \leq 0$

The feasible set contains only x=0

Stationarity would require $1+\lambda(2x)=0$

At x=0 no multiplier can satisfy this equation



A solver choice follows the model

Affine objective and constraints: LP or MILP according to domains

Convex NLP: use convexity with a verified optimality certificate

Nonconvex NLP: a local solver can return different stationary points

Integer decisions plus nonlinear terms require additional methods

Practice A capacity prediction

For the main NLP, change b from 3 to 3.5

Calculate the new point and exact objective value

Predict the change using the old multiplier $\lambda_1=2$

Explain the difference and check the valid interval

Assignment structure

Individual work worth 100 raw points

LP certificate and sensitivity: 25. MILP modeling: 20

Algorithm comparison: 20. Constrained NLP decision: 35

Due December 2, 2026

Assignment evidence and scoring

Submit a PDF report plus a runnable notebook or Python script

Include derivations, required plots, traces and independent checks

Each subpart allocates points to numbers, reasoning and evidence

Equivalent methods and explained numerical tolerances earn credit

Final examination format

90 minutes with 30 single-best-answer questions

Prepare short calculations of directions and one iteration

Practice exact line search, multiplier signs and feasibility

The assignment develops the longer analysis and application

Final retrieval check

What does each KKT condition check?

Why can a constrained minimum have $\nabla f \neq 0$?

Which evidence proves global optimality in today’s example?

When does a multiplier predict only a local value change?

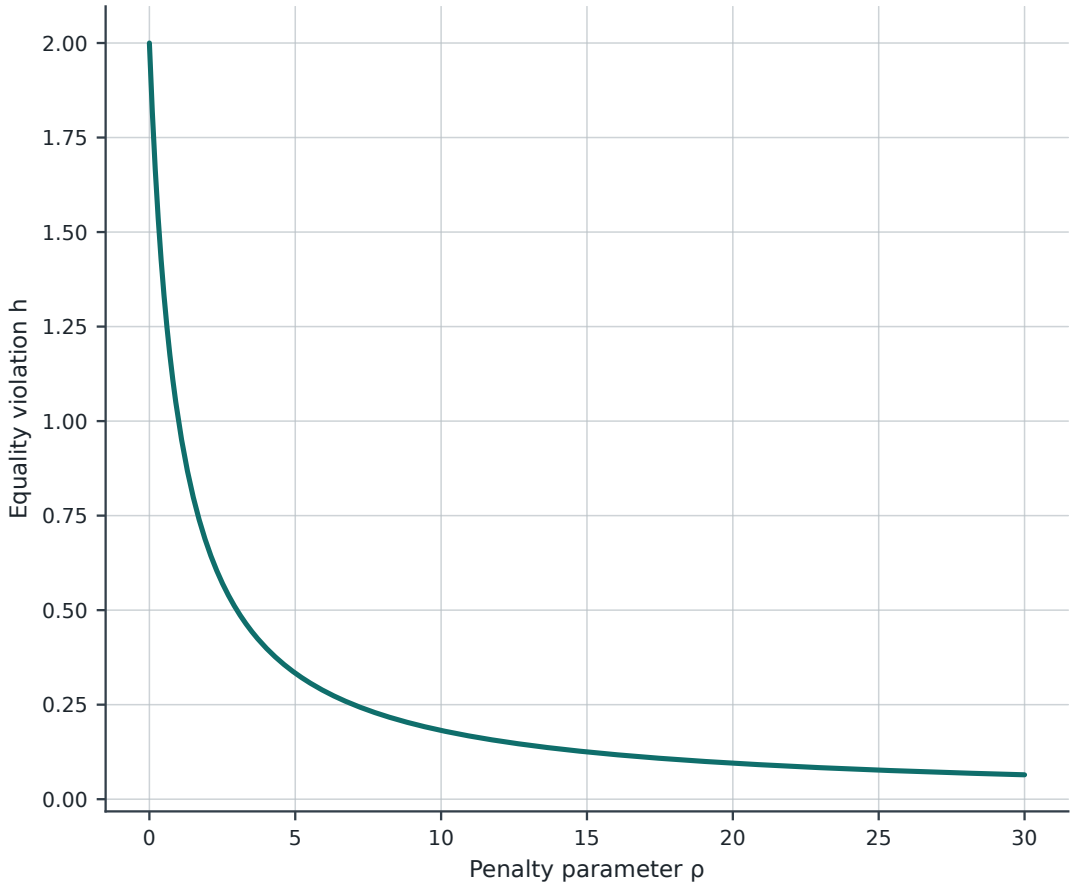
Appendix Quadratic penalty

Equality $h = x + y - 3 = 0$

$P_\rho = f + (\rho/2)h^2$

Stationarity gives $x + y = (5 + 3\rho)/(1 + \rho)$

Violation $h = 2/(1 + \rho)$ persists for every finite ρ



Appendix Augmented Lagrangian

$L_\rho = f + \nu h + (\rho/2) h^2$ for equality $h = 0$

Approximately minimize in x,y with ν fixed

Update $\nu_{k+1} = \nu_k + \rho h(x_{k+1}, y_{k+1})$

Check feasibility and stationarity before stopping

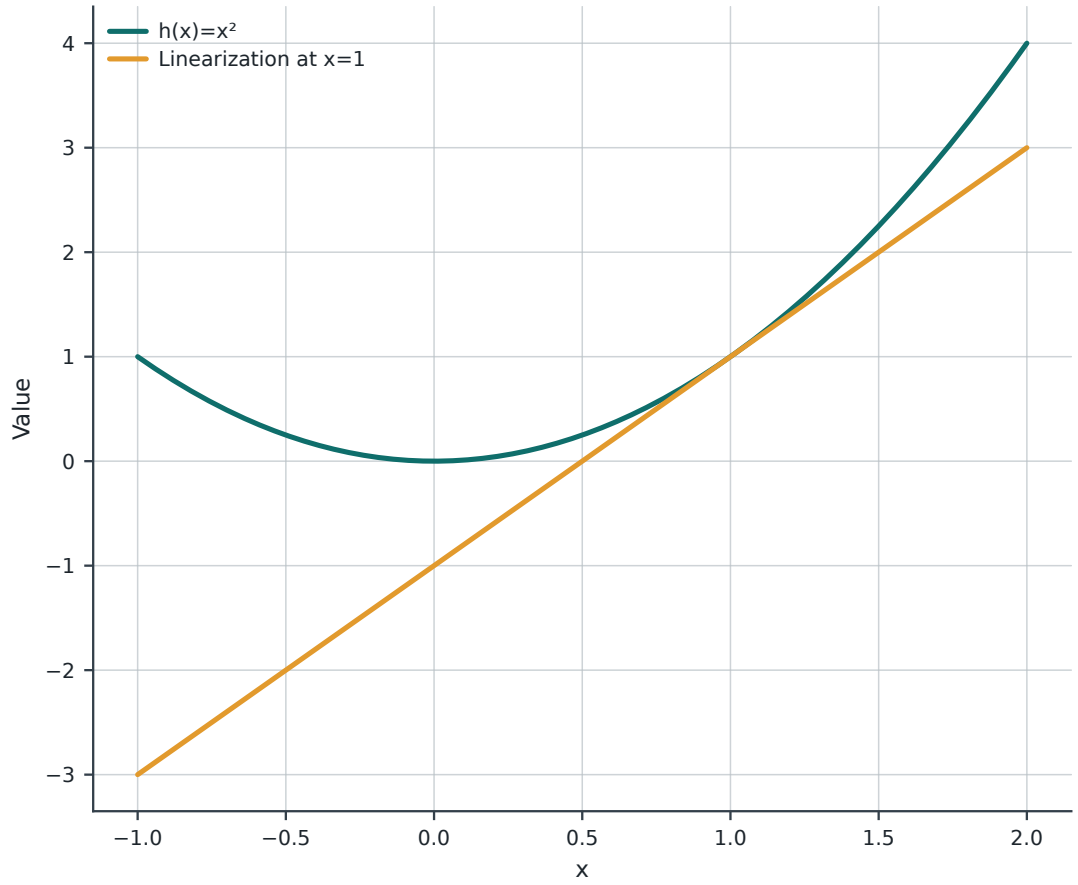
Appendix The SQP subproblem

Use $\nabla f^T d+(1/2)d^T B d$ with $B \approx \nabla^2_{rr} L$

Linearize $h(r)+\nabla h(r)^T d=0$

Linearize $g(r)+\nabla g(r)^T d \leq 0$

Solve the subproblem, control the step and update the model



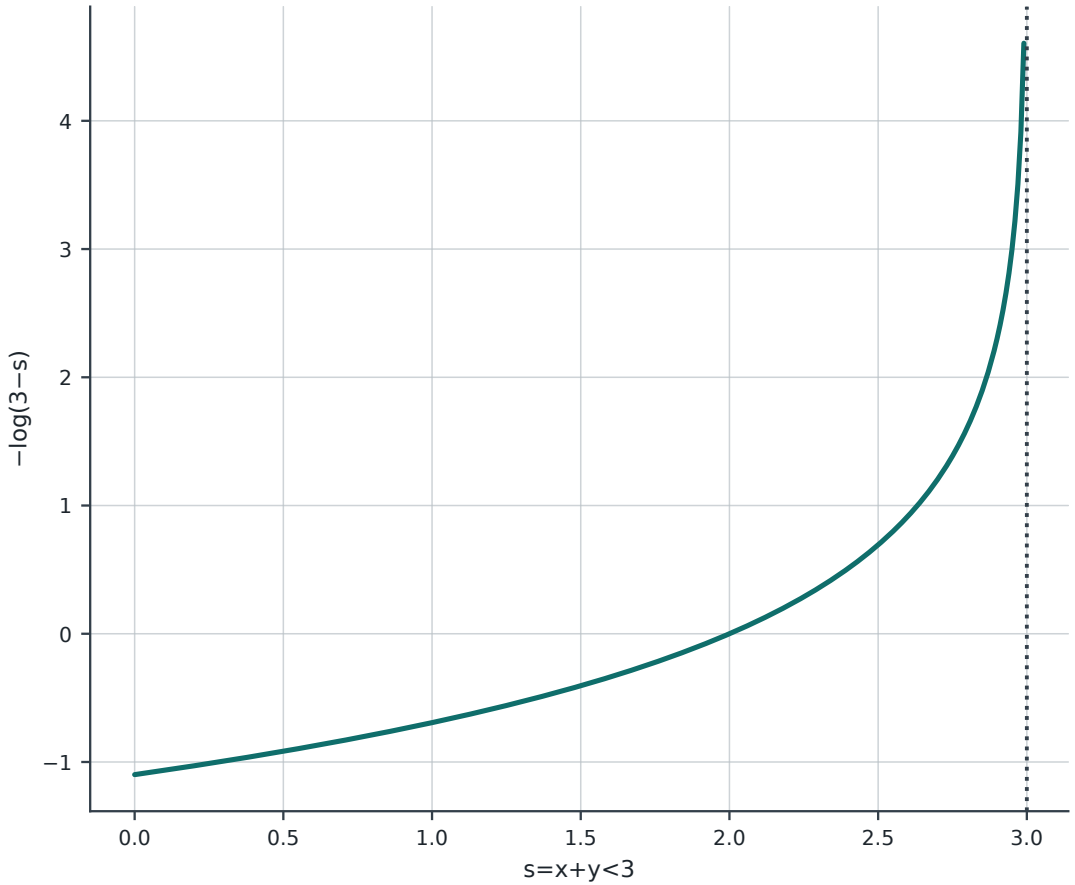
Appendix A barrier term

For $x+y<b$ use $-\mu \log(b-x-y)$

The logarithm requires $b-x-y>0$

As μ decreases, a solution can approach the boundary

Equality constraints still need separate enforcement



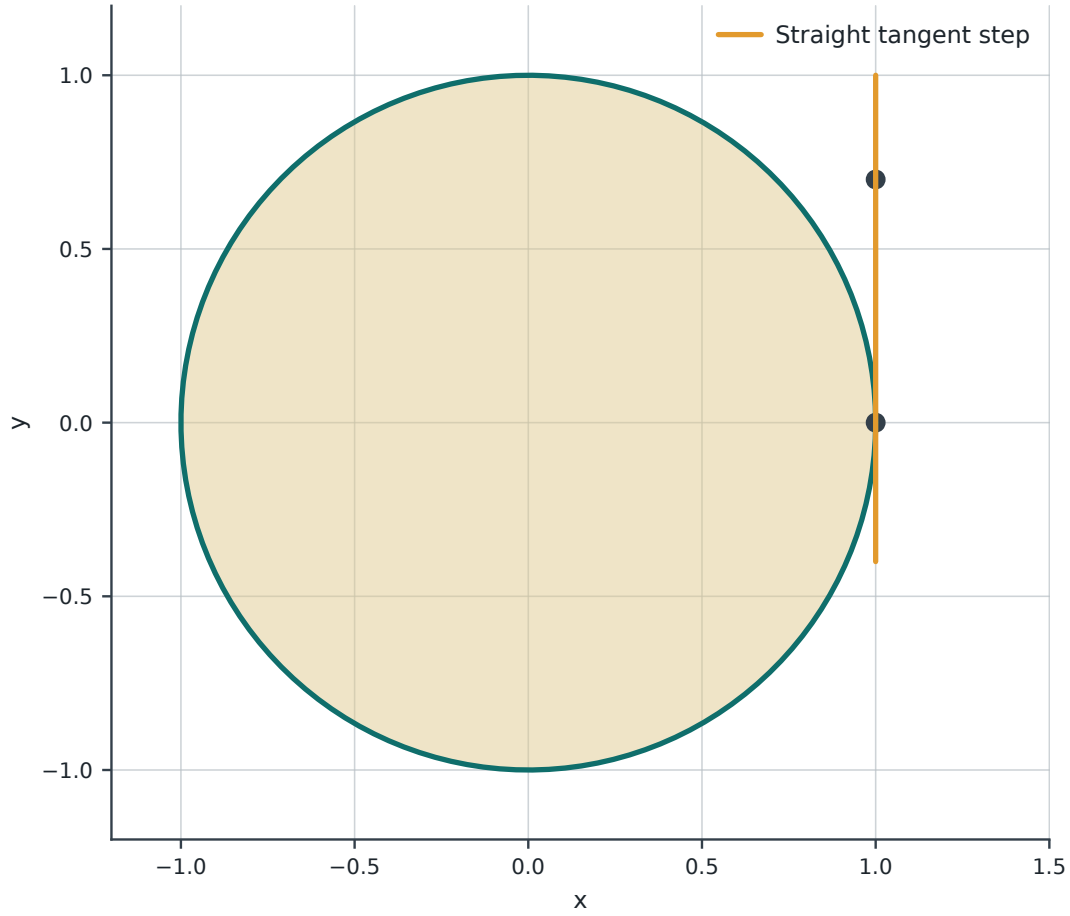
Appendix A tangent need not give a feasible finite step

For $g(x,y)=x^2+y^2-1\le 0$, start at $(1,0)$

The direction $d=(0,1)$ has $\nabla g^T d=0$

But $g((1,0)+td)=t^2>0$ for every $t\neq 0$

Linearized feasibility needs a finite-step feasibility check



Appendix Equality sensitivity

For $x+y=b$, the equality solution is $((b+1)/2,(b-1)/2)$

$\nu=5-b$ and $V(b)=(b-5)^2/2$

With $h=x+y-b$, $dV/db=-\nu$

Equality multipliers need no nonnegative sign restriction

